AGMS: Adversarial Sample Generation-Based Multiscale Siamese Network for Hyperspectral Target Detection

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***Abstract*— Hyperspectral target detection (HTD) has been a critical issue in the field of Earth observation, with widespread applications in both military and civilian domains. However, existing deep learning-based HTD methods are hindered due to insufficient and low-quality prior training samples, as well as inadequate background suppression capabilities. To address these issues, this article proposes an adversarial sample generation-based multiscale Siamese network (AGMS) for HTD. First, the AGMS utilizes the idea of generative adversarial learning based on the prior few targets and diverse backgrounds to generate adversarial target–background sample pairs, thereby producing high-quality training samples, which enhances the distinctiveness between the target and background samples by adversarially training the generator to produce the target/background samples. In addition, to further highlight the targets and suppress the backgrounds, a difference amplification loss and an adaptive weighted binary cross-entropy loss are proposed. Finally, a multiscale convolutional Siamese network model is designed to explore the generated spectral information at multiple levels and achieve target detection through contrastive learning. Numerous experimental results on four real HSI datasets verify the superiority of the AGMS in comparison to many classical and recently proposed HTD methods. The codes are available at https://github.com/ShissHAN/AGMS.**

***Index Terms*— Adversarial learning, deep learning, hyperspectral target detection (HTD), multiscale convolution, Siamese network.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture rich spectral information by providing tens to hundreds of narrow spectral bands from visible to infrared [1], [2], [3]. Hyperspectral target detection (HTD) focuses on identifying and locating specific target pixels of interest from a variety of background pixels, with very rare target prior information. The unique advantages of HTD have facilitated its widespread applications in various fields, including but not limited to military reconnaissance [4], mineral exploration [5], and precision agriculture [6].

HTD involves locating and isolating target pixels in images with limited prior spectra information of targets [7]. In early research, traditional detection methods comprise mainly three categories: spectral matching, subspace models, and statistical distribution hypothesis models. The spectral matching model employed the matched filters to highlight targets while suppressing the background, as demonstrated in [8], a method called the spectral angle mapper (SAM) was proposed, which locates targets by computing the spectral angle between reference target pixels and test pixels. Two typical subspace model-based HTD methods include orthogonal subspace projection (OSP) [9] and target-constrained interference minimization filter (TCIMF) [10]. The statistical distribution hypothesis model usually introduces a certain statistical distribution assumption for targets and backgrounds to construct a generalized likelihood ratio detector, including an adaptive coherence estimator (ACE) [11], an adaptive matched filter (AMF) [12], and a matched subspace detector (MSD) [13]. Furthermore, some scholars proposed the constrained energy minimization (CEM) method [14] to suppress interference. In addition, researchers introduced an iterative weighted CEM method in [15]. In [16], a hierarchical suppression-based technique termed hierarchical CEM (hCEM) was introduced for the detection of constrained energy minimization.

In recent years, some popular machine learning methods such as kernel methods [17], [18], sparse learning [19], [20], [21], low-rank learning [22], and metric learning [23] have been successfully applied to HTD. Kernel methods effectively address linearly inseparable problems by projecting data into high-dimensional latent kernel space, with a typical example termed kernel-constrained energy minimization (KCEM) [24]. Representative methods based on sparse representation include the sparse and joint sparse target detector (SSTD) [25] and the combined sparse and cooperative representation (CSCR) [26] target detector. Methods based on metric learning aim to measure the similarity between samples. For instance, the maximum margin metric learning (MMML) algorithm [27]

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was used for target detection. However, under the limited prior target information, these methods fail to take advantage of deep structural features and local correlation of data.

In recent years, deep learning has attracted attention for its ability to effectively separate targets from backgrounds through abstract feature learning [28], [29], [30]. The deep learning-based methods typically deal with the HTD problem from two perspectives: one is to solve the HTD as a classification problem using massive augmented training samples. For example, in [31], the researchers utilized autoencoders and linear strategies to augment training samples, followed by feature extraction in the convolutional network. Zhu et al. [32] introduced a two-stream convolutional network (TSCNTD) target detector, which adopts a synthetic approach and a sparse classification-based strategy to acquire large-scale training samples. Dong et al. [33] proposed a lightweight convolutional neural network (LCNN-CD), employing joint correlation distance constraints and density peak clustering to enhance the target and background samples. The other perspective adopts unsupervised deep detectors, the core idea of which is to extract discriminative features through unsupervised learning networks [34] and apply them to detect target pixels. For example, Shi et al. [35] presented a novel unsupervised domain adaptive deep learning network with few labeled samples, adversarially adapting similarity/dissimilarity measures from a source domain with sufficient labeled samples to a target domain. In addition, Wang et al. [36] proposed a meta-learning-based Siamese network (MLSN) approach for HTD.

Although the deep learning-based HTD methods have made significant progress, there are still some challenging issues that need to be addressed.

1) Deep models require a large amount of training data to fully realize their advantages. However, in HTD tasks, general training sample enhancement strategies are almost simple and crude and cannot generate the target and background samples with clear distinctiveness.
2) HSIs typically encompass deeply coupled complex and abundant spectral information, which contains different land-cover characteristics under different spectral ranges. Thus, the conventional single-scale feature extraction methodology with a single region often fails to capture these precise detailed features.
3) The core goal of the HTD task is to highlight targets and suppress background interference. However, due to the large amount of background information and very limited prior target information, it presents great challenges to achieve the goal.

To address the above three problems of deep learning-based HTD methods, we propose an adversarial sample generation-based multiscale Siamese network (AGMS) for HTD, with the core ideas of diverse target and background samples generation and target-highlighted and background-suppressed feature learning. The AGMS is composed of an adversarial sample generation network and a multiscale Siamese convolutional network. Specifically, in the sample generation stage, the AGMS adopts the idea of target–background mutual adversarial learning to further capture the latent distribution of real samples, bidirectionally generating more representative target samples and background samples. Subsequently, the generated samples and prior targets are used to train the proposed Siamese network, with a multiscale feature extraction strategy for comprehensively exploring spectral diverse information. Besides, the idea of contrastive learning is introduced to construct pairs in the Siamese network, which can directly address the disparity between positive and negative pairs. Furthermore, to enhance target visibility, suppress background noise, and facilitate learning the correlation between targets and their surroundings, we employ a specialized loss function at each stage. During testing, the testing pixels and the prior target pixels in HSIs are input into the trained multiscale Siamese network for the final detection results. In summary, the novelties and contributions of the proposed AGMS are given as follows.

1) To control the discrimination between the generated targets and backgrounds, we propose an adversarial sample generation (ASG) strategy. The prior background samples and the synthetic target samples are used to construct the target–background adversarial strategy for the generation of target/background samples.
2) To address the issues of insufficient extraction of diverse features in HSIs, we propose a multiscale convolutional Siamese network. This design aims to enhance the model's ability to capture diverse features in hyperspectral data by fusing the features of convolution kernels at different scales.
3) To achieve the suppression of background, we specifically designed a difference amplification (DIA) loss function and an adaptive weighted binary cross-entropy (AW-BCE) loss function, which can empower our model to higher adaptability in a complex environment.

The remainder of the article is organized as follows. In Section II, the proposed AGMS method is introduced in detail. Then, the experiments and the corresponding analysis are presented in Section III. Finally, the conclusions are drawn in IV.

II. Method

In this section, the proposed method is introduced in detail. We first present the overall architecture. Then, the three key components are elaborated, including the adversarial generation of target samples, the reconstruction of background samples, and the multiscale convolutional Siamese network.

A. Overall Architecture

The general framework of the proposed adversarial sample generation-based multiscale Siamese network for target detection is illustrated in Fig. 1.

As illustrated, the proposed method for target detection utilizes the ASG strategy to generate training samples. Initially, the original pixels of HSIs are clustered using target priors to obtain coarse background samples. Subsequently, the target priors and the clustered background samples are separately input into their corresponding encoders and decoders. Using the adversarial mechanism between the real samples

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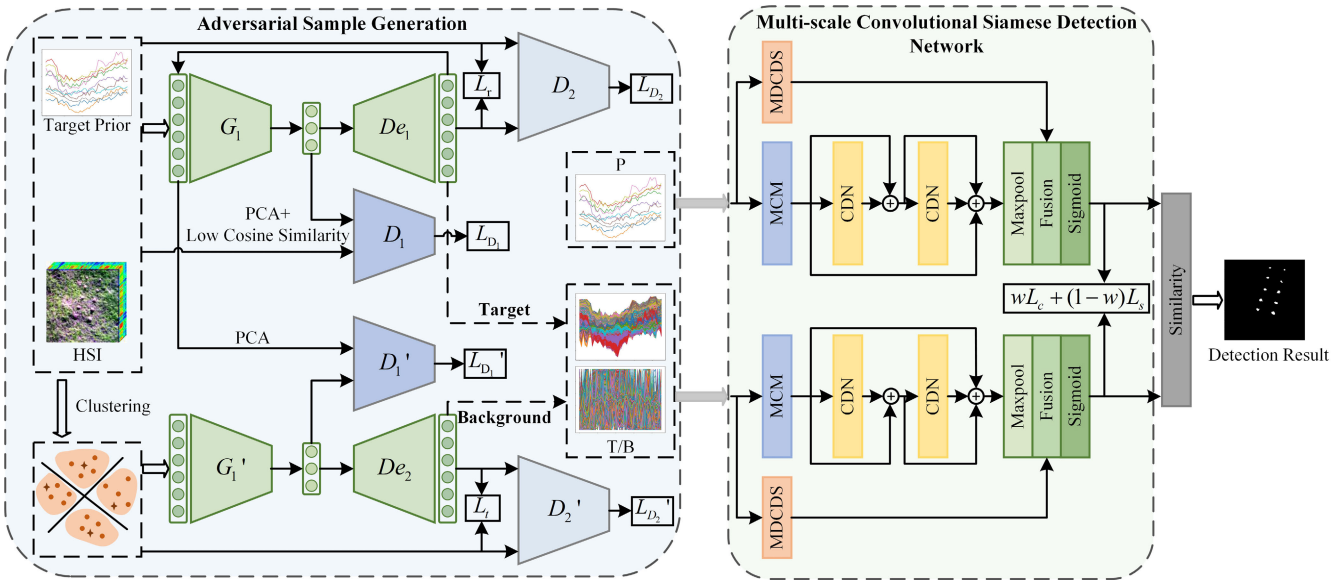


Fig. 1. Flowchart of the proposed AGMS method. It mainly contains two modules: an adversarial sample generation module and a multiscale convolutional Siamese detection network module. G, De, D in the adversarial sample generation module represent the generator, the decoder, and the discriminator, respectively.

and the generated samples through the GAN, the adversarial generation of the target and background samples is performed to construct a more representative set of target samples and background samples, forming the training dataset. After that, the multiscale convolutional Siamese network is established for target detection. The trained target and background samples are fed into the designed multiscale convolutional Siamese network for multiscale feature extraction and fusion. Simultaneously, an AW-BCE loss is devised to guide the training of the network which can efficiently separate target and background pixels. Lastly, the target detection map is achieved by measuring the cosine similarity between the original image and the prior target.

B. Adversarial Generation of Target Samples

A sufficient number of training samples is very critical for a highly efficient HTD system. However, for HTD, no prior information is available other than the very limited spectral signature of targets. Contrastive learning is a machine learning method [37] that emphasizes that similar samples should be close to each other and dissimilar samples should be far apart, which aligns with the contrast between targets and backgrounds in hyperspectral images. Currently, contrastive learning requires positive sample pairs comprising target priors and negative pairs consisting of target priors juxtaposed with the backgrounds. To address the scarcity problem of target samples, we employ adversarial training to generate meaningful target samples, thereby augmenting the dataset for efficient and effective model training.

First, the available target prior t is input into the generator G_1, and then t is mapped to the latent space through an encoder, obtaining its compressed and abstract representation t_z. Subsequently, t_z is fed into the decoder De_1 to generate new target samples t' with similar features reconstructed. To ensure the generation of a sufficient quantity and a variety of target samples, Gaussian noise interference is introduced to the generated target samples t'. These perturbed and diverse samples are then utilized as new target inputs for training, prompting the model to adapt to various forms of interference. This process is repeated until a sufficient number of target samples are generated. Then, the spectra with low cosine similarity to the prior target t are processed using principal component analysis (PCA) to reduce the dimensionality of the background spectra data for matching the dimensionality of t_z. This ensures that the discriminator effectively processes both target and background spectral information. Subsequently, the background spectra x_b and the latent features t_z are fed into the discriminator D_1 for target–background adversarial training. In this process, two generators correspond to two discriminators, and adversarial training is employed to achieve target–background adversarial generation. Specifically, the discriminator D_1 corresponds to the generator G_1, while the discriminator D_2 corresponds to the generator G_2 composed of G_1 and the decoder De_1.

The structure for adversarial generation of target samples is shown in Fig. 2, consisting of two generators (G_1 and G_2) and two discriminators (D_1 and D_2). The main structure of the generator G_1 consists of three fully connected layers. To reduce the sensitivity of the model to the initialization weights, batch normalization (BN) is applied after each fully connected layer. All layers except the output layer are combined with leaky linear rectification (LeakyReLU). G_1 adopts an encoding structure to progressively abstract the features of the input target priors, mapping the data into a latent space to obtain a representation of latent features t_z. The generator G_2 consists of two parts: a generator G_1 and a decoder De_1. The decoder De_1 consists of fully connected layers, BN, and LeakyReLU function activation function, aimed at remapping the encoded low-dimensional features t_z back to the high-dimensional space of the original data to reconstruct

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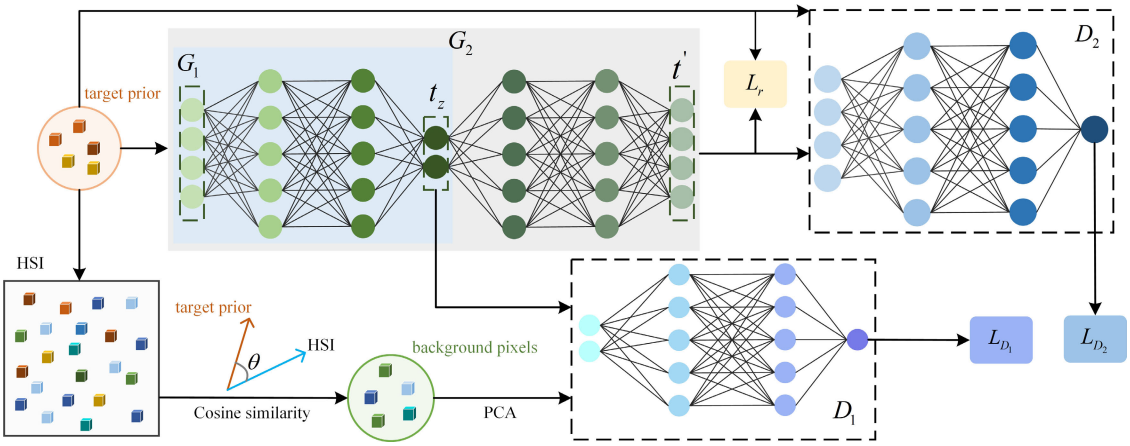


Fig. 2. Flowchart of generating target samples.

the target data t'. Consequently, the discriminators D_1 and D_2 consist of three fully connected layers, with the output of the last fully connected layer passing through a sigmoid activation function, compressing the values within the range of 0–1, representing the probability that the input data are real data.

During the training process, the objective of the generator network G_1 is to extract the latent features t_z of the target spectra. To enhance the quality and diversity of the generated target spectra, ensuring that they better represent the real data distribution, the latent features t_z along with the background spectra are input into the discriminator network D_1. The discriminator D_1 is optimized to maximize discrimination between the target latent features t_z and the background samples. Meanwhile, the feedback from the discriminator drives the generator G_1 to continuously optimize its parameters, thus improving the quality and diversity of the generated samples. The adversarial loss is utilized to guide the optimization of the network

$$L_{D_1} = \min_{G_1}\max_{D_1}\left\{E_{x_b\sim p(x_b)}\left[\log D_1(x_b)\right] + E_{t\sim p(t)}\left[\log(1-D_1(G_1(t)))\right]\right\}. \tag{1}$$

The objective of the generator network G_2 is to generate samples similar to real target spectra to deceive the discriminator network D_2. Meanwhile, the objective of the discriminator network D_2 is to distinguish between the target spectra generated by the generator G_2 and the real target spectra. Therefore, a dynamic adversarial training paradigm is established between the generator G_2 and the discriminator D_2, and the adversarial loss between G_2 and D_2 can be mathematically formulated as follows:

$$L_{D_2} = \min_{G_2}\max_{D_2}\left\{E_{t\sim p(t)}\left[\log D_2(t)\right] + E_{t\sim p(t)}\left[\log(1-D_2(G_2(t)))\right]\right\}. \tag{2}$$

In addition, to generate more representative and high-quality target samples, the network computes the difference between the generated output and the target samples and utilizes the mean square error loss to quantify this difference. By minimizing this loss function, the generator network G_2 adjusts its parameters to approximate its output, making it closer to the input target samples

$$\min_{t'} L_r = \min_{t'} |t - t'||_2^2. \tag{3}$$

Ultimately, through adversarial training, the generator G_2 can generate target samples t' that are representative and distinctly different from the background samples.

C. Background Sample Reconstruction and Purification

Although predetection or clustering algorithms can estimate the background information of HSIs to some extent, the backgrounds selected by these methods may still contain sparse targets and noise, which will reduce the performance of subsequent detection networks. Therefore, by employing the ASG strategy to the initial background samples selected by the clustering algorithms, purer background samples can be reconstructed. With the mechanism, adversarial reconstruction gradually enhances the authenticity and purity of the generated background samples through the adversarial process between the generator and the discriminator. The structure for the adversarial reconstruction of background samples is similar to the adversarial generation of targets, as shown in Fig. 1, consisting of generators G_1' and G_2' and discriminators D_1' and D_2'. However, it should be noted that although the overall structure is similar, the inputs and outputs of the generator and the discriminator are not the same. First, the initial background samples b obtained by clustering the original image HSI are inputted into the generator G_1', resulting in the latent feature representation b_z of background samples. Subsequently, b_z is entered into the discriminator D_1'. The additional input t'' of D_1' is obtained by performing PCA on the generated target samples. Then, the latent representation of background samples b_z is inputted into the decoder De_2 for decoding, obtaining the reconstructed background data b'. Finally, the reconstructed background data b', along with the original background samples, are inputted into the discriminator D_2' for evaluation.

During the training process, the generator network G_1' strives to produce background latent feature representations b_z that can deceive the discriminator D_1', while the discriminator

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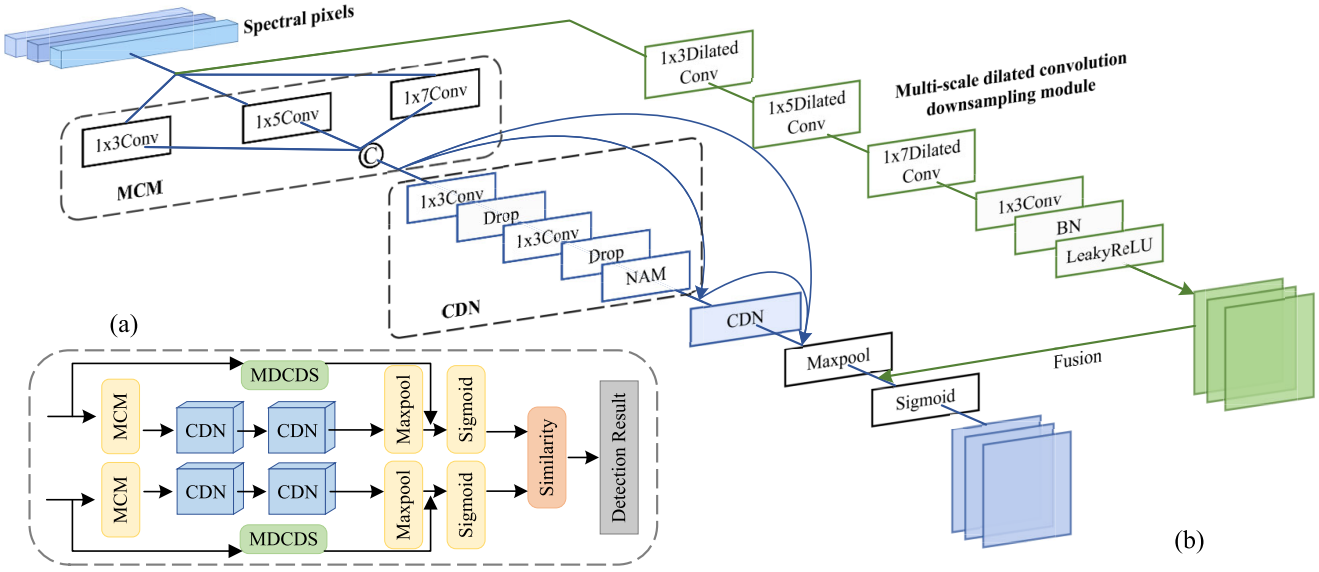


Fig. 3. Flowchart of the multiscale convolutional Siamese network. (a) Siamese network architecture diagram. (b) Branch-specific module components.

D_1' strives to distinguish between the real target samples and the generated background feature representation. This adversarial training mechanism helps to improve the performance of the generator network, enabling it to learn more accurate and effective background feature representation, thereby improving the highlighting of targets and the suppression effect of backgrounds. In this process, the adversarial loss is applied to drive the optimization problem

$$L'_{D_1} = \min_{G_1'} \max_{D_1'} \left\{ E_{t'' \sim p(t'')} \left[\log D_1'(t'') \right] + E_{b \sim p(b)} \left[\log\left(1 - D_1'\left(G_1'(b)\right)\right) \right] \right\}. \tag{4}$$

By learning to generate realistic background samples, the generator network G_2' aims to produce background spectral samples that can deceive the discriminator network D_2', making it difficult to distinguish between real and generated samples. At the same time, the discriminator network D_2' effectively distinguishes between the background spectral samples generated by the generator network G_2' and the real background spectral samples, ensuring the quality and authenticity of the generated samples. Therefore, a dynamic adversarial training process is established between the generator network G_2' and the discriminator network D_2', where the adversarial loss between G_2' and D_2' can be expressed as follows:

$$L'_{D_2} = \min_{G_2'} \max_{D_2'} \left\{ E_{b \sim p(b)} \left[\log D_2'(b) \right] + E_{b \sim p(b)} \left[\log\left(1 - D_2'\left(G_2'(b)\right)\right) \right] \right\}. \tag{5}$$

To increase the discrimination between the reconstructed background samples and the target prior and ensure that the reconstructed samples can truly characterize the backgrounds, we design the reconstruction loss L_r' and the DIA loss L_a. Here, L_r' aims to minimize the error between the input data b and the reconstructed data b' to the fullest extent

$$\min_{b'} L_r' = \min_{b'} ||b - b'||_2^2. \tag{6}$$

On the other hand, the DIA loss L_a is designed to increase the dissimilarity between the reconstructed background samples and the target priors, thereby prompting the model to learn more distinct features for distinguishing between the targets and backgrounds. We use N to denote the number of samples of the reconstructed background and calculate the cosine similarity between each reconstructed background spectrum b_i' and the prior target t. A higher value indicates greater similarity to the prior target. To amplify the difference between the target and the reconstructed background and reduce the similarity, the average similarity is used as the DIA loss

$$L_a = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{b_i' t}{\left(\|b_i'\|\right)\left(\|t\|\right)} \right]. \tag{7}$$

The total loss function can be represented as

$$L_t = L_r' + L_a. \tag{8}$$

In the end, through adversarial training, the generator G_2' can produce background samples b' that are purer and significantly different from the target samples.

D. Multiscale Convolutional Siamese Network

The proposed multiscale convolutional Siamese network is illustrated in Fig. 3(a), which mainly consists of the multiscale convolution module (MCM), the convolutional feature extraction module (CDN), and the multiscale dilated convolution downsampling module (MDCDS).

1) MCM: As shown in Fig. 3(b), the MCM aims to address the complexity of HSI data and the inherent challenges in target detection. By incorporating convolutional modules of three distinct scales (1×3, 1×5, and 1×7), the objective is to effectively capture features that represent the detail characteristics from different scales in HSIs. Functionally, the MCM module enhances feature representation by harnessing information across different spatial scales. Varying

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kernel-sized convolutional layers capture different ranges of context in hyperspectral data. Through the concatenation of features from differing scales, the module efficiently integrates multiscale information while preserving critical details vital for target detection tasks. Furthermore, each convolutional layer is followed by a BN and a LeakyReLU, which stabilizes training and enhances feature learning by alleviating gradient explosion/vanishing issues. In summary, the MCM module plays a pivotal role in extracting robust and informative features from hyperspectral data, thereby improving the performance of HTD.

2) CDN: As shown in Fig. 3(b), the proposed CDN module aims to further refine the feature extraction process after the MCM and enhance the feature representation capability. This is achieved through a series of components designed to capture local spectral information, reduce overfitting, and focus on relevant features. Specifically, the CDN module consists of two 1×3 1-D convolutional modules, each with 64 neurons, a stride of 1, and padding of 1. These 1-D convolutional modules help in capturing local spectral information from the input data. Two adaptive drop block modules (AdapDrop) [38] are included to effectively reduce overfitting and improve the model's generalization ability. In addition, one normalization-based attention module (NAM) assists in better focusing on important features and reducing the impact of irrelevant information. To progressively deepen feature representation, two consecutive CDN modules are employed, enhancing the abstraction and generalization capabilities of the input data.

3) MDCDS: As clearly depicted in Fig. 3(b), the MDCDS module is designed to enhance the capture of contextual information at different scales, improve the model's discriminative power for multiscale features, and mitigate the problem of attenuation of spectrally critical features that may occur during multiple convolution operations. The MDCDS module consists of a series of three different scale dilated convolutional modules concatenated with a downsampling module. Each dilated convolutional module sets the number of neurons, convolutional kernel size, and stride to 64, 1×3, and 1, respectively. To ensure output size, the padding is set to 1, 3, and 5, and the dilation rates are set to 0, 3, and 5, respectively. The purpose of these dilated convolutional modules is to capture contextual information at different scales, thereby improving the model's ability to discriminate multiscale features. The downsampling module comprises a 1-D convolution with the number of neurons, convolutional kernel size, and stride set to 64, 1×3, and 3, followed by a BN layer and a LeakyReLU activation function. The goal of downsampling is to reduce the spatial dimension of the feature maps while preserving important feature representation, providing a broader space for information exchange among the subsequent modules for feature fusion. Overall, the MDCDS module effectively captures contextual information at different scales and enhances the model's discriminative power for multiscale features, while mitigating the attenuation of spectrally critical features.

The overall framework of one subnetwork in the multiscale convolutional Siamese network is illustrated in Fig. 3(b). The spectral pixels undergo feature extraction through a multiscale convolutional module and two CDN modules. Across three interconnected modules, each output seamlessly transitions as input for the next, to enrich and enhance features while maximizing the retention of original spectral information. This approach guarantees a thorough feature extraction while minimizing spectral information loss. Due to the vast amount of feature information extracted by convolutions, the spatial dimension reduction of feature maps is achieved through max-pooling layers, which retain the most significant feature within local regions, thereby achieving feature compression. Simultaneously, the original spectral pixels undergo MDCDS to obtain efficient spectral feature representation while preserving semantic information and spatial context relationships in the output. The two results are then fused in addition to enabling the model to effectively capture information in the spectral data. Finally, the output is mapped to a range between 0 and 1 for subsequent detection.

During the training process, the two inputs of the Siamese network are the prior target spectrum and the generated target/background samples. If the input for the upper branch is fixed as the target prior, and the input for the lower branch is the target, the label for the training sample is set to 1; otherwise, it is set to 0. In addition, we keep the same number of positive and negative samples.

In the proposed network, there are two loss functions. The first is the contrastive loss function L_c [39], where for input sample pairs (z_0, z_i), the extracted feature representations of the network are $f(z_0)$, $f(z_i)$. We use the idea of increasing the distance between negative samples and decreasing the distance between positive samples to guide model training with the following loss function:

$$L_c = y||f(\mathbf{z}_0) - f(\mathbf{z}_i)||_2^2 + (1-y)\max\left(0, m - ||f(\mathbf{z}_0) - f(\mathbf{z}_i)||_2^2\right)^2 \quad (9)$$

where y represents the label of the sample pairs, $f(\cdot)$ denotes the network output values, and m is a hyperparameter that indicates the degree of distance for negative sample pairs.

In addition, to enhance the background suppression ability of the network, an AW-BCE loss is proposed. The nonlinear function $q(x)$ is utilized to suppress backgrounds while highlighting the target pixels. As the similarity of positive sample pairs is greater than that of negative sample pairs, $q(x)$ can maintain higher values for target pixels and produce lower values for background pixels, which can effectively suppress the backgrounds. The exponential suppression function is defined as

$$q(x) = 1 - e^{-\lambda x}. \quad (10)$$

The BCE loss function is formulated as follows:

$$L_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[d_i \cdot \log(l_i) + (1-d_i)\cdot \log(1-l_i)\right]. \quad (11)$$

The AW-BCE loss function L_s is defined as follows:

$$L_s = \frac{1}{N}\sum_{i=1}^{N} k_i \cdot \mathrm{BCE}(d_i, l_i) \quad (12)$$

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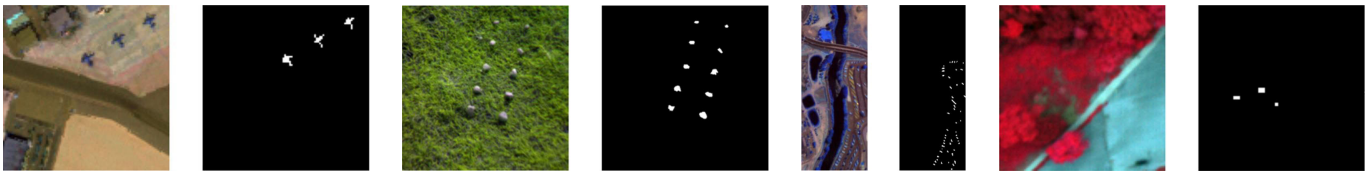


Fig. 4. Dataset image scenes and ground truth. (a) AVIRIS. (b) Nuance-CRI. (c) RenoUrban. (d) Viareggio.

where N is the number of samples, k_i is the weight of the ith sample, d_i is the distance of the ith sample pair, and l_i is the label of the ith sample. The calculation of k_i is as follows:

$$k_i = 1 - e^{-\lambda d_i} \tag{13}$$

where λ is a hyperparameter set to 10, the value of k_i is within the range [0, 1], representing the weight of the corresponding sample. Finally, BCE(d_i, l_i) denotes the binary cross-entropy loss function, which sums the weighted loss of all samples and then divides the sum value by the number of samples to obtain the final loss value.

The total loss function comprises two components: contrastive loss L_c and AW-BCE loss L_s

$$L = wL_c + (1 - w)L_s \tag{14}$$

where w is a weighting parameter to balance the influence of each component on the overall computation.

III. Experiments

A. Experimental Details

1) Data Description: Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) San Diego dataset: The first dataset is the airport data of San Diego, CA, USA, acquired by the AVIRIS sensor, which contains the aircraft, housetop, and airport runway. The experimental data image size is 100×100, the spectral resolution is 10 nm, the spatial resolution is 3.5 m, and the band range is 370–2510 nm. The water vapor absorption bands in the wavelength range of 1350–1420 and 1810–1940 nm are removed, including 1–6, 33–35, 97, 107–113, 153–166, and 221–224 bands, with the other 189 bands preserved. The aircraft was selected as occupies 0.0058 of the HSI. The false color images and ground-truth maps are shown in Fig. 4(a).

Nuance CRI Grass-Stone dataset: The second dataset is a real HSI acquired by Nuance's CRI hyperspectral imager, a handheld imager with a spectral resolution of about 10 nm in 46 bands and a spectral range of 650–1100 nm. The entire image is 400×400 pixels and contains ten human-arranged stone targets of interest and several background features (e.g., bare soil and grass). The stone is selected as the target of interest in the experiment, and it occupies 0.007825 of the HSI. Due to the different sizes and materials of the target stones, as well as the solar altitude angle, which causes the stones to produce some shadows, it can cause more difficulties in stone detection. Fig. 4(b) shows the false color image scene and the location of the target in this dataset.

Reno Urban data: In addition, the third dataset is a scene from Reno, Nevada, USA, which contains 356 spectral bands. After removing the low-quality bands, there are 287 bands left. A region of 250×100 pixels is chosen as the test data. The cars-covering pixels at the bottom-right of the image are the targets. Due to the limitation of target size and spatial resolution, many of them are subpixel targets. The tenth band of the hyperspectral data and the truth map of Reno data are shown in Fig. 4(c).

Viareggio data: The fourth dataset is the Viareggio scene, containing 511 spectral bands. The selected image depicts a facility near a parking lot. A 100×100 pixel area is selected for the test data and the three green panels on the lower left of the image are the targets, which are difficult to separate from the backgrounds because the targets are distributed near the green grass. The true image of the hyperspectral data and the truth map of the Viareggio data are shown in Fig. 4(d).

2) Evaluation Criterion: To analyze the performance of the proposed AGMS detection, three qualitative and quantitative evaluation metrics were utilized: receiver-operating characteristic (ROC) curve, area under the curve (AUC) value, and separability plots.

The ROC curve is a commonly used tool for assessing the performance of binary classification models. The plotted image shows the false positive rate (FPR) on the horizontal axis and the true positive rate (TPR) on the vertical axis, with FPR labeled as P_F and TPR labeled as P_D. With a given τ, P_D and P_F can be calculated by

$$P_D = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad P_F = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{15}$$

where TP denotes the number of pixels that are correctly detected as targets at a given threshold τ, FN represents the number of pixels that incorrectly detect targets as backgrounds at a given threshold τ, FP denotes the number of pixels that incorrectly detect backgrounds as targets at a given threshold τ, and TN denotes the number of pixels that are correctly detected as backgrounds at a given threshold τ. Generally, the closer to the upper left corner the ROC curve is, the better the performance of the classifier is, while the closer the curve to the diagonal line is, the worse the performance of the classifier is.

The AUC value is one of the important metrics for evaluating the performance of a classification model. The AUC value represents the area under the ROC curve, ranging from 0 to 1. Specifically, the classical $\text{AUC}_{(P_D, P_F)}$ represents the area under the ROC curve of P_D and P_F, and the higher the area value is, the better the effectiveness of the detector is. $\text{AUC}_{(\tau, P_F)}$

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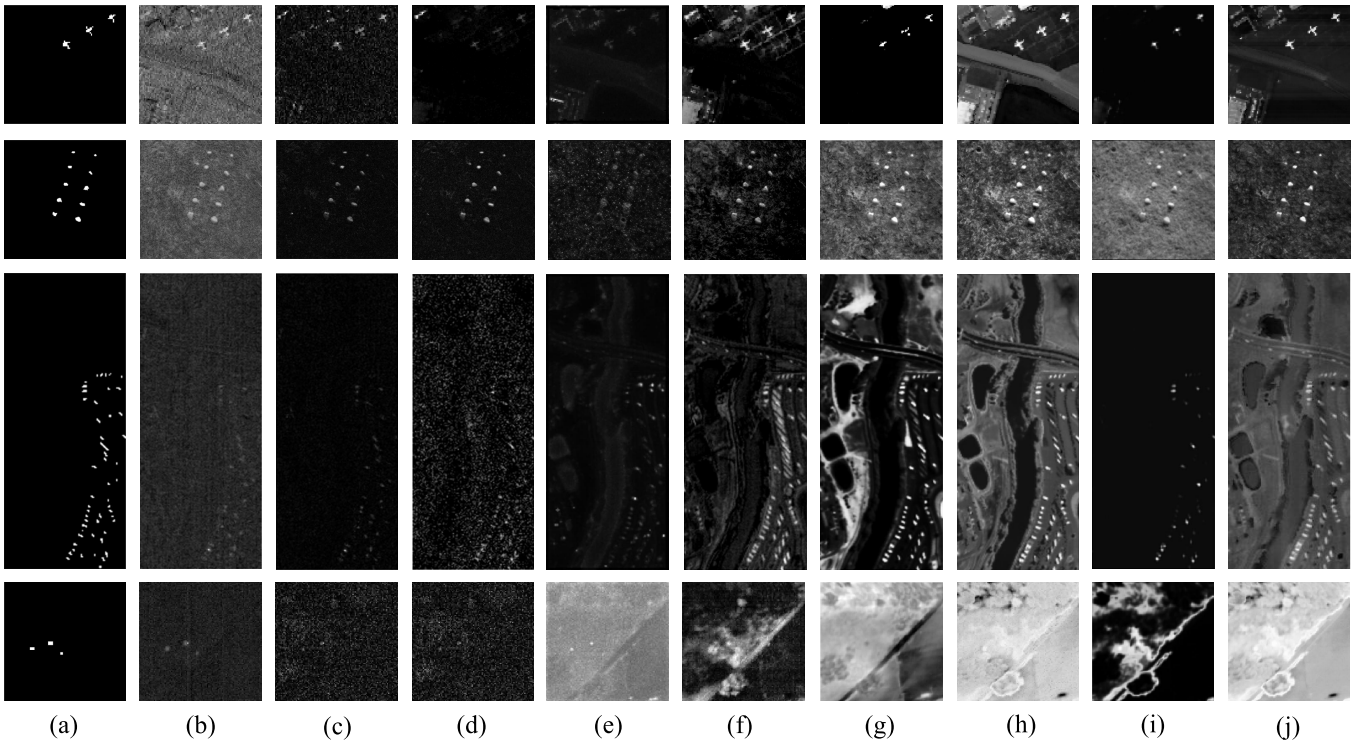


Fig. 5. Detection maps for different methods. (a) Ground truth. (b) ACE. (c) CEM. (d) HCEM. (e) CSCR. (f) DSC. (g) TSCNTD. (h) LCNN-CD. (i) HTD-IRN. (j) AGMS.

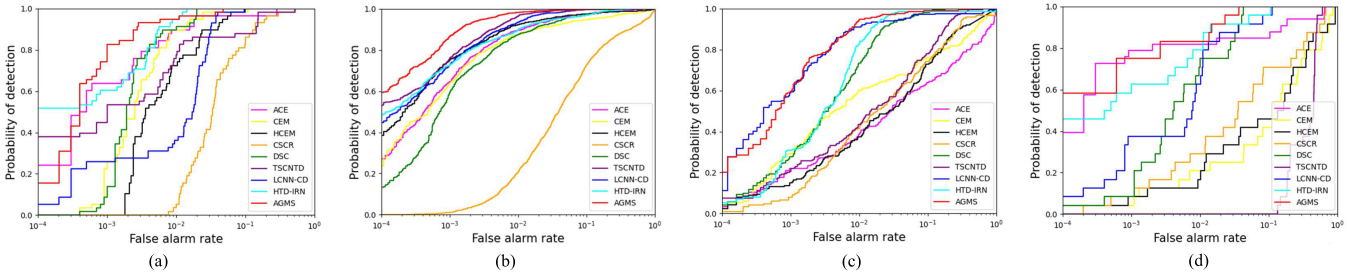


Fig. 6. ROC curves for all the detectors on the four datasets. (a) AVIRIS. (b) Nuance-CRI. (c) RenoUrban. (d) Viareggio.

represents the area under the ROC curve of τ and P_F, and the lower the area value is, the better the background suppression performance is. Furthermore, an overall AUC value is calculated by comprehensively considering $\mathrm{AUC}_{(P_D,P_F)}$ and $\mathrm{AUC}_{(\tau,P_F)}$:

$$A_{DF} = \mathrm{AUC}_{(P_D,P_F)}/\mathrm{AUC}_{(\tau,P_F)}. \tag{16}$$

The separability plot [40] is a statistics-based method to visualize the separability between positive and negative samples in binary hypothesis testing problems. This graphical representation relies on box plots of different colors to represent the data distributions of the positive and negative samples separately. To observe the position and range of the box plot, the distinguishability between the targets and the backgrounds is intuitively revealed. The median is located in the center of the box plot, the quartiles define the top and bottom boundaries of the box plot, and the whiskers show the range of data excluding outliers. This graph not only visually demonstrates the distribution of data but also directly reflects the level of separability between the targets and backgrounds by comparing the overlap of box plots.

3) Experimental Settings: To validate the detection performance of the proposed AGMS, we compared it with eight other detection methods, including three classical detectors (ACE [11], CEM [14], and HCEM [16]), two machine learning methods (CSCR [26] and DSC [41]), as well as three latest deep learning methods (TSCNTD [32], LCNN-CD [33], and HTD-IRN [42]). It is worth mentioning that, for a fair comparison, all comparison methods used optimal parameters determined through experiments and the referenced literature.

For the proposed AGMS method, the learning rates for pretraining the generator and the discriminator were set to 1e-3 and 1e-4, respectively. The number of epochs and batch size were set to 5000 and the total number of samples, respectively. The learning rate, the number of epochs, and batch size for the multiscale convolutional Siamese network were set to 1e-3, 50, and 64, respectively, and the Adam optimizer was used. All the experiments were conducted by using an NVIDIA

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TABLE I
AUC VALUES ON THE FOUR DATASETS

$AUC_{(P_D,P_F)}\uparrow$	AVIRIS	Nuance-CRI	RenoUrban	Viareggio
ACE	0.9877	0.9893	0.8032	0.9787
CEM	0.9941	0.9739	0.8841	0.7652
HCEM	0.9882	0.9869	0.8917	0.8147
CSCR	0.9411	0.8712	0.9028	0.8548
DSC	0.9963	0.9905	0.9895	0.9784
TSCNTD	0.9738	0.9981	0.9419	0.6335
LCNN-CD	0.9861	0.9971	0.9782	0.9852
HTD-IRN	0.9974	0.9955	0.9806	0.9909
AGMS	**0.9978**	**0.9991**	**0.9946**	**0.9959**

TABLE II
AUC VALUES ON THE FOUR DATASETS

$AUC_{(\tau,P_F)}\downarrow$	AVIRIS	Nuance-CRI	RenoUrban	Viareggio
ACE	0.3623	0.3096	0.1522	0.1406
CEM	0.0789	0.0478	0.0252	0.1033
HCEM	0.0172	0.0601	0.0674	0.1097
CSCR	0.0586	0.0914	0.0471	0.4781
DSC	0.2172	0.1191	0.1049	**0.0242**
TSCNTD	0.0033	0.1049	0.1649	0.1053
LCNN-CD	0.1389	0.0812	0.0894	0.0845
HTD-IRN	0.0431	0.4029	0.0498	0.3723
AGMS	**0.0011**	**0.0005**	**0.0008**	0.0582

GeForce RTX 3090 graphics processing unit (GPU) and 16 GB of random access memory (RAM).

B. Detection Performance

Fig. 5 visualizes the detection results of the AGMS and the other comparison methods. It is not difficult to see from the results that AEM and CSCR could only detect partial targets on the four datasets and fail to effectively suppress the backgrounds. CEM, HCEM, and HTD-IRN exhibit strong background suppression across all four datasets but only detect some targets without a significant salient effect. DSC, TSCNTD, and LCNN-CD effectively detect most targets but struggle to suppress the backgrounds well, occasionally misclassifying some backgrounds as targets, especially on the Reno Urban dataset. Although HTD-IRN performs well on the Nuance-Cri and Viareggio datasets, it misses some targets on the other datasets. The AGMS demonstrates strong background suppression capability on the Nuance-Cri and AVIRIS datasets. The Reno Urban and Viareggio datasets correctly identify targets, but their background suppression capability needs improvement. Overall, our method exhibits more stable detection rates across the four datasets, with most target pixels correctly identified.

Fig. 6 shows the ROC curves, where the red curve represents the proposed AGMS. It can be seen that the AGMS consistently achieves the best ROC curves across all four datasets, with higher detection probabilities and lower false alarm rates. For the Viareggio dataset, when the false alarm rate is less than 0.001, the ROC curve of the ACE detector is slightly higher than that of the AGMS. This could be attributed to the similarity between the spectral signatures of the green board targets and the grass backgrounds in the Viareggio dataset.

TABLE III
AUC VALUES ON THE FOUR DATASETS

$A_{DF}\uparrow$	AVIRIS	Nuance-CRI	RenoUrban	Viareggio
ACE	2.73	3.20	5.28	6.96
CEM	12.60	20.37	35.08	7.41
HCEM	57.45	16.42	13.23	7.43
CSCR	16.06	9.53	19.17	1.79
DSC	4.59	8.32	9.43	**40.43**
TSCNTD	295.09	9.51	5.71	6.02
LCNN-CD	7.10	12.28	10.94	11.66
HTD-IRN	23.14	2.47	19.69	2.66
AGMS	**907.10**	**1998.20**	**1243.25**	17.11

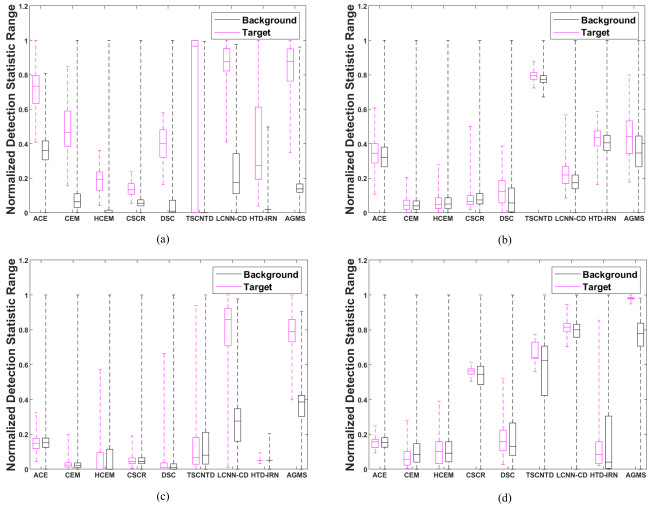


Fig. 7. Maps of target background separation for all detection methods on the four datasets: (a) AVIRIS. (b) Nuance-CRI. (c) RenoUrban. (d) Viareggio.

The AUC values for all the methods on the four datasets are given in Tables I and II, with $AUC_{(P_D,P_F)}$ reflecting detection accuracy and $AUC_{(\tau,P_F)}$ responding to background suppression accuracy. $\uparrow$ means the higher the better and the $\downarrow$ means the lower the better. The optimal results are shown in bold and the suboptimal results are underlined. Comparing the $AUC_{(P_D,P_F)}$ values of each method, for the AVIRIS and Nuance-Cri datasets, DSC, HTD-IRN, and LCNN-CD achieve relatively stable $AUC_{(P_D,P_F)}$ values, with all exceeding 0.9. However, the $AUC_{(P_D,P_F)}$ values of the AGMS method are higher, consistently exceeding 0.99 on all four datasets, indicating its superior detection performance. For $AUC_{(\tau,P_F)}$ of the AGMS method, the minimum was obtained on three datasets, with the remaining one dataset being the next smallest position. Similarly, our AGMS method can yield the highest A_{DF} values on the AVIRIS, Nuance-Cri, and Reno Urban datasets as shown in Table III. Overall, the AUC values of the comparative detectors perform better on the first two datasets, which may be attributed to the higher quality of some commonly used datasets.

The separability plots of each method on the four datasets are shown in Fig. 7, aiming to intuitively validate the effectiveness of our method in separating targets from backgrounds. For almost all the datasets, compared with the other methods, our method consistently exhibits larger gaps between the magenta boxes (representing targets) and the black boxes

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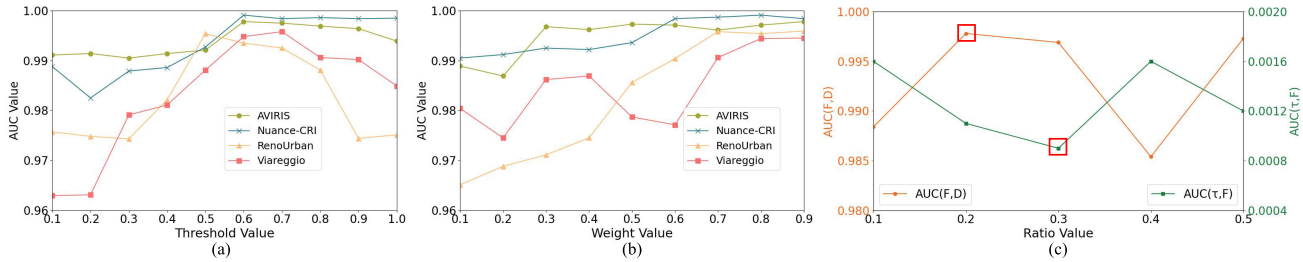


Fig. 8. Parametric analysis chart. (a) Threshold m. (b) Weight w. (c) Sample ratio. The ratio of the number of generated sample parameters was experimentally analyzed on the AVIRIS dataset.

(representing backgrounds), indicating a better target and background separability ability. In addition, the central lines of the magenta boxes for our method are generally positioned higher, suggesting that the AGMS can better highlight targets. In particular, for the Viareggio dataset, the other comparative methods are unable to effectively separate target and background boxes, whereas the AGMS can achieve a significantly large gap between targets and backgrounds. For the AVIRIS, Reno Urban, and Viareggio datasets, the black-box area (representing the background) from the AGMS method is smaller, indicating that the method can suppress the backgrounds.

C. Parameter Analysis and Ablation Study

1) Parameter Analysis: This section analyzes three crucial parameters of the proposed AGMS, including the threshold parameter m controlling the degree of target–background separation in the contrast loss function, the weight parameter w of the Siamese network loss function, and the proportion of adversarial generated samples.

The analysis of the threshold m is depicted in Fig. 8(a). The role of the threshold m is to regulate the intensity of the contrast loss function. When m is small, the intensity is low and the model focuses more on the distance between the sample and the positive sample. In contrast, when m is larger, the intensity is higher and the model focuses more on the distance between the samples and the negative samples. m was set based on the similarity of the positive and negative sample pairs within the set of candidate parameters ([0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]). Five consecutive runs were conducted, and the average AUC values for the four datasets are presented in the figure. The AVIRIS and Nuance-Cri datasets exhibit minimal fluctuations across the entire range of m. The trend of curves generally shows an initial increase followed by a decrease, with relatively better performance when m is between 0.5 and 0.7. Considering the overall analysis of the four datasets, we chose 0.6 as the threshold for the intensity of the contrast loss function.

During the training phase of the Siamese network, two loss functions are employed: the contrast loss function L_c and the AW-BCE loss function L_s. We adopt a weighted fusion approach for training the loss function, hence we should consider the weight w. The range of w is set to ([0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]). The average AUC values on the four datasets are illustrated in Fig. 8(b). The overall trend of the curves generally shows an initial increase followed by stable performance, with relatively better performance when m is between 0.8 and 0.9. Comparing the datasets, the Viareggio dataset exhibits greater fluctuations under different weight parameters, possibly due to its sensitivity to model optimization or the presence of more noise or interference in the data. A certain degree of AW-BCE loss function can effectively suppress background noise and stabilize detection performance. Therefore, we chose 0.9 as the threshold for the contrast loss function.

The training samples generated by the ASG strategy need to possess both fidelity and variability. Excess variability may compromise fidelity, while excessive fidelity may hinder diversity. Therefore, it is crucial to find a balance between them. To address this, we set the ratio of the number of generated samples as [10%, 20%, 30%, 40%, 50%] of the total number of pixels in HSIs. Subsequently, we conducted experiments on the AVIRIS dataset and observed the variation in performance in terms of AUC values with respect to the number of generated samples, as illustrated in Fig. 8(c). $\mathrm{AUC}_{(P_D,P_F)}$ and $\mathrm{AUC}_{(\tau,P_F)}$ reached their optimal values with a sample ratio of 20% and 30%, respectively, and both the values were worse when the ratio was set to 40%. When the sample size reaches a certain level, its diversity may be significant. However, as the number of samples increases further, the fidelity of the data might be affected due to potential overfitting or the inclusion of less informative samples. Another important factor is computational cost. At a 30% sample ratio, although the $\mathrm{AUC}_{(\tau,P_F)}$ metric is slightly improved compared to 20%, the increase in computational resources required is substantial. This additional computational load does not yield a proportional improvement in performance, making the 30% ratio less efficient compared to the 20% ratio. Therefore, we chose a 20% ratio of the number of generated samples, ensuring a good balance between fidelity, diversity, and computational efficiency.

2) Ablation Study: This section analyzes the necessity and effectiveness of the ASG strategy, the multiscale convolution (MSC) strategy, and the AW-BCE loss in the proposed method. These modules are compared with partial sample generation (target generation TG or background generation BG), simple convolution strategy, and BCE loss. During the training sample acquisition stage, coarse training samples can be obtained through background clustering, target prior, and background simple synthesis, instead of adversarially generated training samples. During the training stage, the option to add a multiscale module and whether to optimize the network using

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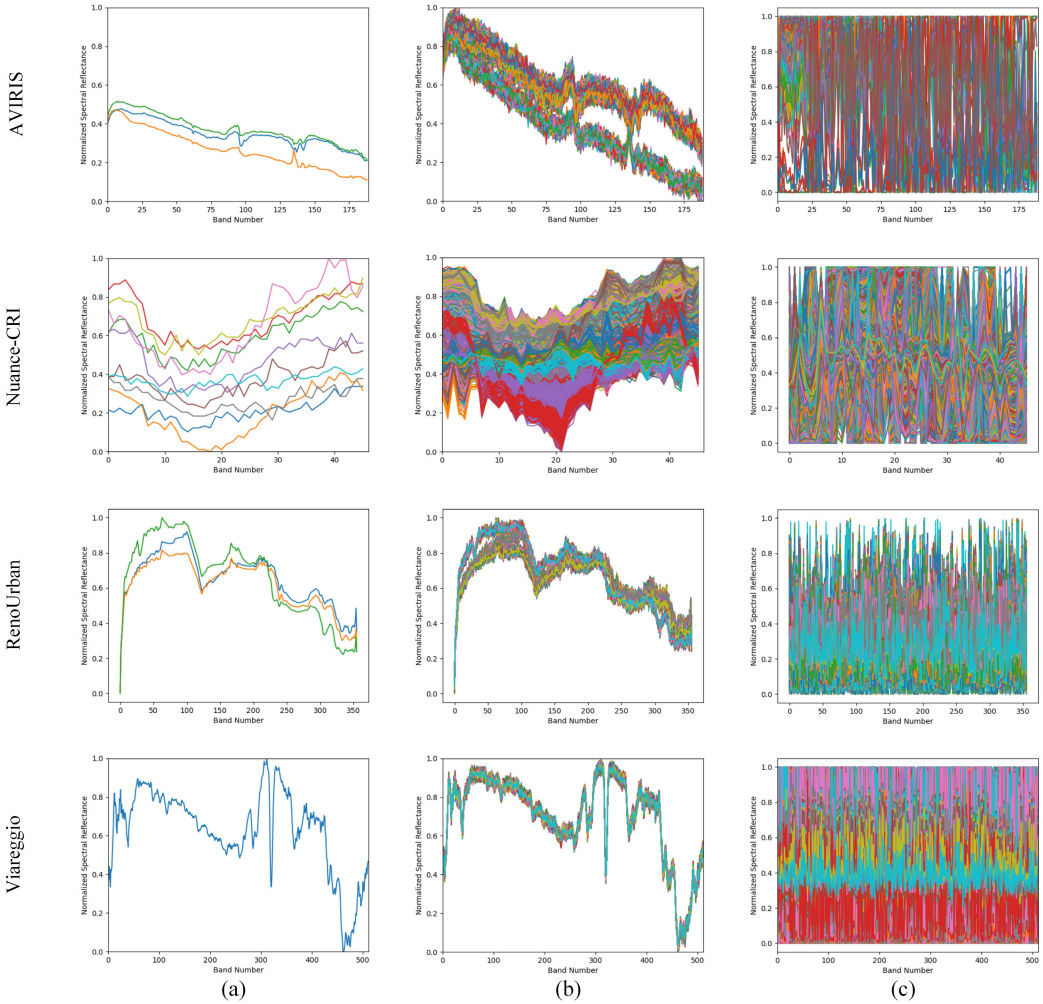


Fig. 9. Visualization of sample spectra curves on the four real HSIs. (a) Prior target spectra. (b) Augmented target prior spectra. (c) Background samples.

BCE loss or AW-BCE loss can be chosen. Different choices for training sample generation, detection network modules, and loss functions result in multiple comparative experiments, labeled ASG-BCE, MSC-AWBCE, BG-MSC-AWBCE, TG-MSC-AWBCE, ASG-MSC-BCE, and ASG-MSC-AWBCE.

The average AUC values on the four datasets are shown in Table IV.

1) ASG Strategy: Compared with BR-MSC-AWBCE, TR-MSC-AWBCE, and MSC-AWBCE, the comparative experiments of ASG-MSC-AWBCE show a significant increase in AUC values on the four datasets. This indicates that the ASG strategy effectively improves the quality of target and background training samples, which in turn leads to better detection performance.

2) MSC Strategy: The comparison between ASG-BCE and ASG-MSC-BCE indicates that the MSC strategy plays a significant role in improving the detection performance for each dataset. The MSC strategy captures features at different scales, which helps in addressing the issue of scale variation in targets, leading to improved detection accuracy.

3) AW-BCE Loss Function: The experimental results of ASG-MSC-BCE suggest that the AW-BCE loss function contributes to improving detection accuracy, particularly on the Reno Urban dataset.

TABLE IV
AUC SCORES OF THE ABLATION STUDY ON DIFFERENT DATASETS

Methods	AVIRIS	Nuance-CRI	RenoUrban	Viareggio
ASG-BCE	0.9871	0.9856	0.9776	0.9798
MSC-AWBCE	0.9772	0.9858	0.9743	0.9818
BR-MSC-AWBCE	0.9949	0.9885	0.9746	0.9682
TR-MSC-AWBCE	0.9778	0.9958	0.9799	0.9816
ASG-MSC-BCE	0.9915	0.9935	0.9821	0.9891
ASG-MSC-AWBCE	**0.9978**	**0.9991**	**0.9945**	**0.9959**

In summary, these experimental results demonstrate the effectiveness of the ASG strategy, MSC strategy, and AW-BCE loss function in enhancing detection performance. The combination of these components significantly outperforms the baseline methods, highlighting the importance of each component in the proposed approach.

D. Sample Visualization

To visually demonstrate the quality of the generated samples by the proposed AGMS method, we visualized the spectral

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curves of the target and background samples on the four HSI datasets. Fig. 9 presents the visualization results of target priors, generated target samples, and reconstructed background samples from the AGMS method on the four real HSI datasets. It can be observed from Fig. 9(b) that the generated target samples largely preserve the spectral curve profile of the target priors, with some spectral differences among the target samples. The difference can bring about more useful spectral variability, which is beneficial to enhance the generalization and detection capabilities of our proposed method. In addition, Fig. 9(c) reveals that the generated background samples exhibit a distinct curve profile compared with the target samples, indicating that the generated background samples are more discriminative. From these qualitative visualization experiments, it can be seen that the proposed method can effectively generate the target and background samples with good fidelity, diversity, and discrimination, which can facilitate the training of deep detection networks.

IV. Conclusion

This article has proposed a novel HTD method named AGMS. Specifically, to address the dilemma of insufficient and low-quality training samples, the AGMS proposed to generate realistic and diverse training samples through adversarial training with enhanced target–background discrimination. In addition, to tackle the background suppression problem, the AGMS designed background suppression loss functions, namely the DIA loss and the AW-BCE loss, to prepare training samples and train the Siamese network, respectively. Finally, a multiscale convolutional Siamese network module was employed for target detection. Extensive experiments were conducted on four challenging hyperspectral datasets to validate the superiority of the AGMS method over several state-of-the-art approaches.

However, the migration performance of our method is low and requires retraining in the face of new TD tasks. Subsequent research will focus on how to improve the method's portability and increase the efficiency of TD tasks.

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